

STAR FOX 64

PRACTICE ROM

PLAYER GUIDE v0.5

Save states * Frame advance * Hitbox visualizers * Free camera * Input macros



WELCOME, BETA TESTER

Star Fox 64 Practice ROM * Beta Release

Thank you sincerely for testing out the beta version of the Star Fox 64 Practice ROM. This ROM attempts to maintain as much vanilla code as possible while unlocking helpful practice tools similar to those found in other N64 practice ROMs. I want it to feel like Star Fox 64 - just with extra tools layered on top, not replacing anything you already know.

This early on, there are bound to be bugs and crashes. If you run into anything - a freeze, a wrong value, a menu that behaves unexpectedly - please send it over and I will get a fix released as soon as possible. Every report helps make the ROM better for everyone.

RELEASES

sageraces.com

BETA

staging.sageraces.com

REPORTING BUGS

Please include: what level you were on, what you were doing when it crashed or broke, and whether you can reproduce it. Screenshots or video are a big help. Anything you share makes the next version better.

SD CARD - NOT WORKING YET

The SD card save/load feature is under active development and does NOT currently work on EverDrive cartridges. Do not rely on it to save your work. RAM save slots (D-Left / D-Right) work fine. SD card support will be enabled in a future release once it is stable.

This guide covers the controls and features included in this beta build. If something in the ROM does not match what is described here, that is a bug - please report it.

QUICK REFERENCE - CONTROLS

DURING GAMEPLAY (radial menu closed)

INPUT	ACTION
Z + D-Right	Open the practice radial menu
D-Down	Pause / unpause the game (frame advance)
D-Up	Step one frame forward (while paused)
D-Left	Quick-save to the active slot
D-Right	Quick-load from the active slot

WHILE RADIAL MENU IS OPEN

INPUT	ACTION
Analog Stick + A	Navigate and select menu options
B	Go back / close the menu
L (alone)	Cycle active save slot backward
R (alone)	Cycle active save slot forward
Z	Save to SD card (NOT WORKING YET on EverDrive)
Z + B	Load from SD card (NOT WORKING YET on EverDrive)
Hold START (~1.5 sec)	Return to the title screen

NOTE

D-pad shortcuts (save, load, pause, frame step) are ONLY active during normal gameplay. They do not fire while the radial menu is open.
L and R slot-cycle work only when no D-pad button is held simultaneously.

RADIAL MENU NAVIGATION

Press Z + D-Right during gameplay to open the practice radial menu. Tilt the analog stick toward an option to highlight it, then press A to select. Press B to close or go back. Some options open a second radial layer.



RESTART	Instantly restart the current level.	LEVELS	Open the level select screen.
DISPLAY	Open the display submenu (overlays, camera, macro).	LOADOUT	Configure gear before launching a level.
SAVE	Save your state to the active slot.	CHEATS	Toggle gameplay cheats.
LOAD	Load state from the active slot.	SD CARD	SD card saves (NOT WORKING YET on EverDrive).

SAVE STATES

The practice ROM supports up to 4 RAM save slots (requires Expansion Pak). Snapshots capture your exact position, velocity, enemy state, charge level, phase progress, and all game flags - everything needed to resume precisely.

DURING GAMEPLAY (menu closed)

D-Left	Quick-save to the active slot (also works while paused)
D-Right	Quick-load from the active slot

WHILE RADIAL MENU IS OPEN

Menu > SAVE	Save via radial menu (same as D-Left during gameplay)
Menu > LOAD	Load via radial menu (same as D-Right during gameplay)
L (alone)	Cycle slot backward (slot 1 -> 4 -> 3 -> ...)
R (alone)	Cycle slot forward (slot 1 -> 2 -> 3 -> ...)
Z	Save to SD card (NOT WORKING YET on EverDrive)
Z + B	Load from SD card (NOT WORKING YET on EverDrive)

TIP

If you save while the game is paused (frame-advance mode), loading that slot restores the paused state automatically - great for practising a single frame window repeatedly.

LEVEL SELECT

Open with Menu > LEVELS. Use the analog stick or D-pad to browse all levels and pick a checkpoint phase. Press A to launch. The Boss Test option (Corneria Carrier) lets you practice the end-game boss fight from any position.

NOTE

The Loadout screen (Menu > LOADOUT) is separate from level select. Configure your gear first, then pick a level to launch with those settings.

LOADOUT

Configure everything about your starting state before launching a level. Settings persist between launches until you change them.

LASER	SINGLE / TWIN / HYPER (default: HYPER)
BOMBS	Starting bomb count
LIVES	Starting life count
GOLD RINGS	Starting gold ring count
HEALTH	SHORT (normal) or LONG (silver Arwing)
RIGHT / LEFT WING	NONE / BROKEN / INTACT
FALCO / SLIPPY / PEPPY	ALIVE or DOWN
EXPERT	Toggle expert mode on/off
HIT COUNT	Starting hit count (useful for route planning)
PREV PLANETS	Mark which planets are already cleared (affects route-dependent stage unlocks such as Sector Z)

CHEATS

Toggle cheats from Menu > CHEATS. Active cheats show a coloured indicator label in the HUD. All cheats persist until manually turned off.

AUTO CS

Fires a charge shot at the earliest possible frame (timer > 10). Useful for verifying charge-shot windows without manual timing.

INF HP

Shields never deplete - you cannot die.

INF BOMB

Bomb count stays at 9.

INF LIFE

Lives stay at 99.

INF BST

Boost meter never depletes - boost freely without penalty.

DISPLAY SUBMENU

Open with Menu > DISPLAY (point stick NE, press A). A second radial layer appears with 7 options arranged around the screen.

SKIP CUTS - Automatic cutscene skip

When enabled, all in-level cutscenes are skipped automatically. Toggle on/off from the Display radial. Active by default.

INPUTS - Live input overlay

Draws a live overlay of your current controller inputs on screen - buttons, stick position, triggers. Useful for tutorial recording or reviewing your own inputs frame by frame alongside frame-advance.

CAMERA - Free camera mode**NOTE**

You must pause the game (D-Down) before enabling Free Camera. The CAMERA option is greyed out in the menu when the game is running.

Detaches the camera from Fox so you can fly freely through the level. Useful for studying geometry, hitboxes, and spawn zones.

Analog Stick	Look (yaw/pitch)
C-Up / C-Down	Move camera forward / backward
C-Left / C-Right	Strafe camera left / right
R (hold)	Move faster (5x boost)
D-Left / D-Right	Cycle placement object (None > Bomb > Explosion)
A (no object)	Toggle the camera HUD overlay
A (with object)	Place a range-visualizer marker at the aim point
Z	Toggle hitbox rendering on/off
B	Exit free camera and return to normal view

TIP

Use D-Left/D-Right to select a Bomb or Explosion object, then press A to drop a range-visualizer marker at the crosshair. The ring shows the blast radius - helpful for learning charge-shot and bomb positioning.

STATS - HUD stats overlay

Opens a sub-list where individual stat counters can be toggled on/off:

HUD OVERLAY	Master toggle for the entire stats bar
LAG FRAMES	Count of lag frames accumulated this level
SPEED	Current moving speed (base + boost combined)
CHARGE TIMING	After firing a charge shot, shows how many frames early or late you released relative to the ideal window
MISSED INPUTS	Tracks how many input windows were missed
HIT TRACKING	Tracks hit-count events (direct hits, bonuses, despawns)

CS METER - Charge shot charge bar

Adds a visual progress bar below the reticle showing how charged your shot is. The bar fills from 0 to 100% as the charge timer climbs from 0 to 10 - hitting 100% is the first frame you can release for a charge shot. The bar stays solid once lock-on search activates (timer > 20).

VISUALIZERS

Open via Display > VISUALS. Renders coloured geometric overlays on top of the game world to help study hitboxes and actor spawn triggers.

Hitbox options

HITBOXES	Master toggle - enables the hitbox rendering system
ACTORS	Draw hitboxes around enemy actors
SCENERY	Draw hitboxes around destructible scenery
ITEMS	Draw hitboxes around collectibles (rings, bombs, upgrades)
PLAYER	Draw the player Arwing hitbox
FLASH	Flash actors bright white when they take a hit

Spawn zone options

SPAWN ZONES	Master toggle - show actor spawn trigger volumes
SPAWN ACTORS	Highlight enemy spawn triggers
SPAWN ITEMS	Highlight item spawn triggers
SPAWN SCENERY	Highlight scenery spawn triggers

ENEMY HP DISPLAY

Shows live HP numbers floating above enemies. Accessed via Visualizers > ENEMY HP. Options:

SHOW	ON / OFF - master toggle for enemy HP numbers
SORT	NEAREST (closest enemy first) or HIGH HP (descending HP)
MIN HP	Only show enemies at or above this HP threshold (0 = all, options: 0 / 1 / 5 / 10 / 25 / 50)
FILTER	ALL enemies, or BOSSES ONLY
MODELS	Toggle rendering of enemy 3D models on/off - hide to reduce visual clutter while still seeing HP numbers

TIP

Setting FILTER to BOSSES ONLY and hiding enemy MODELS is useful during boss fights - you see the boss HP bar without cluttering the screen.

INPUT MACRO - Record and Replay

The macro system records your controller inputs frame-by-frame and plays them back deterministically. Open via Display > MACRO.

RECORD	Press to start recording; press again to stop. Inputs are captured from the current game position.
PLAY	Replay the recorded inputs. If BIND STATE is on, playback first restores the bound save state.
REWIND	Jump the playback head back to frame 0.
TRIM	Discard everything after the current playback position.
FRAMES	Displays the total number of recorded frames (read-only).
BIND STATE	Link the macro to the active save state. Playback will auto-load that state before replaying.
LOOP	Continuously replay the macro from the start.

TIP

Workflow: save your position (D-Left), open Macro > RECORD, perform the inputs you want to study, stop recording, then enable BIND STATE and press PLAY. The macro will reload your save state and replay the exact same inputs on each loop - perfect for frame-perfect technique practice.

SD CARD SAVES

NOT WORKING YET ON EVERDRIVE

SD card saving is in active development and does NOT currently work on EverDrive cartridges. Do not use it to store saves you care about - it may crash or silently fail. This section documents the intended behaviour once the feature ships in a stable release.

Once finished, the SD CARD feature will let you persist snapshots across sessions to a FAT-formatted SD card (targeting EverDrive-64 X7/X8). RAM save slots are lost when the console is powered off - SD saves will be how you keep setups permanently.

SD SAVE

Name and write the current state to the SD card. A simple on-screen keyboard lets you enter a file name.

SD LOAD

Browse saved snapshots on the card and load one. Use the stick/D-pad to scroll, A to select, B to cancel.

WHEN IT IS READY

Watch sageraces.com and staging.sageraces.com for updates. SD card support will be announced when it is stable and tested on hardware.

FRAME ADVANCE - DETAILED REFERENCE

Frame advance lets you step through gameplay one frame at a time - invaluable for studying hitbox interactions, input windows, and timing.

D-Down	Toggle pause on/off
D-Up (tap)	If running: pause. If already paused: step one frame
D-Up (hold)	After a short delay (~12 frames), auto-step at 2-frame cadence

NOTE

While paused, buttons you hold are re-injected as fresh presses on each stepped frame. This means holding A while stepping will fire lasers or accumulate charge shot charge, just as it would during normal play.

TIPS & TRICKS

CHARGE SHOT WINDOW PRACTICE

Enable CS METER from Display > CS METER. The bar reaches full (100%) at the exact frame you can release for a charge shot. Enable CHARGE TIMING (Stats overlay) to see whether you released early or late, measured in frames.

STUDYING A SPECIFIC SECTION

Save at the start of the section (D-Left), play through it, then quick-load (D-Right) to return instantly. No menus needed - you can repeat the segment as fast as you can press D-Right.

COMPARING ROUTES

Use multiple save slots (L/R to cycle) to save checkpoints at different points and measure route times against each other.

HITBOX VERIFICATION

Pause the game, enable Free Camera and Hitboxes (VISUALS), then use the camera to inspect hitbox geometry from any angle. Place a bomb-range marker (D-Right, then A) to see whether enemies fall within charge-shot blast radius.

BOSS PRACTICE

Use Menu > LEVELS > BOSS TEST to jump straight into Corneria Carrier boss. Combine with INF HP cheat to study attack patterns without dying.

RETURN TO TITLE QUICKLY

Hold START for about 1.5 seconds. The game fades to the title screen. Useful for resetting a run without powering off.

SF64 Practice ROM v0.5 | github.com/qwoodmansee/sf64-practice-rom | For questions, visit the SF64 Speedrunning Discord